



YEAR 12 EXTENSION 1 MATHEMATICS ASSESSMENT JUNE 2009

TIME : 70 MINUTES

NAME		RESULT	
DIRECTIONS	- Full working should be shown in every question. Marks may be deducted for careless or badly arranged work. - Use black or blue pen only (<i>not pencils</i>) to write your solutions. - No liquid paper is to be used. If a correction is to be made, one line is to be ruled through the incorrect answer.		
QUESTION 1.	Differentiate (a) $\tan^{-1} 4x$ (b) $\sin^{-1}(e^{2x})$		2 2
QUESTION 2.	Find the following : (a) $\int \frac{dx}{9+25x^2}$ (b) $\int_0^{\frac{1}{\sqrt{3}}} \frac{dx}{\sqrt{1-3x^2}}$		2 3
QUESTION 3.	Prove that $\frac{d^2x}{dt^2} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$ where v is the velocity of a particle at time t.		2
QUESTION 4.	(a) Sketch $y = 2 \sin^{-1} \frac{x}{3}$ stating the domain and range. (b) Find the area between the curve and the y-axis in the first quadrant.		3 3

QUESTION 5.	Without the use of a calculator, evaluate the following showing all working: a) $\cos^{-1}\left(-\frac{1}{2}\right)$ b) $\tan^{-1}\left(\tan\frac{2\pi}{3}\right)$ c) $\sin\left[\tan^{-1}\left(-\frac{2}{3}\right) + \cos^{-1}\left(\frac{12}{13}\right)\right]$	2 2 3
QUESTION 6.	Evaluate $\int_0^2 \sqrt{4-x^2} dx$ using the substitution $x = 2 \sin \theta$.	4
QUESTION 7.	Find $\int \frac{(\ln x)^2}{x} dx$ using the substitution $u = \ln x$.	3
QUESTION 8.	A particle oscillates back and forth in a straight line with velocity v (ms^{-1}) in position x where $v^2 = 16x - 4x^2 + 20$. a) Prove that the motion is simple harmonic motion. b) Find the extremities of the particle's motion. c) Find the period of the motion. d) Find the maximum speed reached by the particle.	3 2 1 2
QUESTION 9.	A particle is moving along x-axis, starting from a position 2metres to the right of the origin with an initial velocity of $5ms^{-1}$ and an acceleration given by $a = 2x^3 + 2x$. a) Show that $\dot{x} = x^2 + 1$ b) Hence find an expression for x in terms of time t.	2 3
QUESTION 10.	A ball is projected from the top of a building 20 metres high with initial velocity 15 ms^{-1} and an angle of projection θ to the horizontal. The ball just clears a wall which is 6.25 metres high and 30 metres away from the foot of the building. Let $g = 10ms^{-2}$. a) Derive the equations for the horizontal and vertical displacement. b) Find the two possible angles of projection c) Find the exact time to reach the wall for the smaller angle of projection.	2 3 2

Answers Fr. 12, 3rd Unit, June 2009 Total (51)

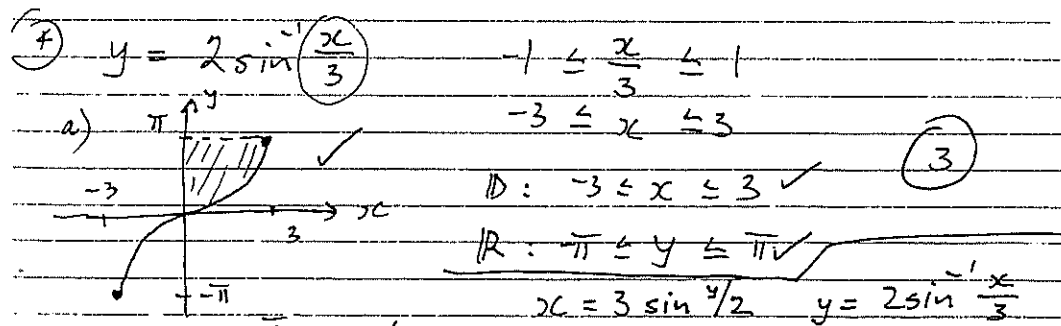
① a) $\frac{d}{dx} \tan^{-1} 4x = \frac{4}{1+(4x)^2} = \frac{4}{1+16x^2}$ (2)

b) $\frac{d}{dx} (\sin^{-1} e^{2x}) = \frac{2e^{2x}}{\sqrt{1-(e^{2x})^2}} = \frac{2e^{2x}}{\sqrt{1-e^{4x}}}$ (2)

② a) $\int \frac{dx}{9+25x^2} = \int \frac{dx}{25(\frac{9}{25}+x^2)} = \frac{1}{25} \cdot \frac{1}{\frac{3}{5}} \cdot \tan^{-1} \frac{x}{\frac{3}{5}} = \frac{5}{3} \cdot \frac{1}{25} \tan^{-1} \frac{5x}{3} + C = \frac{1}{25} \tan^{-1} \frac{5x}{3} + C$ (2)

b) $\int_0^{1/\sqrt{3}} \frac{dx}{\sqrt{1-3x^2}} = \int_0^{1/\sqrt{3}} \frac{dx}{\sqrt{3} \sqrt{\frac{1}{3}-x^2}} = \frac{1}{\sqrt{3}} \left[\sin^{-1} \sqrt{3}x \right]_0^{1/\sqrt{3}} = \frac{1}{\sqrt{3}} \left[\frac{\pi}{2} - 0 \right] = \frac{\pi}{2\sqrt{3}}$ (3)

③ $\frac{d^2x}{dt^2} = x = \frac{dv}{dt} = \frac{dv}{dx} \cdot \left(\frac{dx}{dt} \right) = v \cdot \frac{dv}{dx}$ (since $\frac{d}{dv} \left(\frac{1}{2}v^2 \right) = v$)
 $= \frac{dv}{dx} \cdot \frac{d}{dv} \left(\frac{1}{2}v^2 \right) = \frac{d}{dx} \left(\frac{1}{2}v^2 \right)$ (2)



b) $A = - \int -3 \sin \frac{y}{2} dy = \left[-6 \cos \frac{y}{2} \right]_{\pi}^{\pi} = \frac{y}{2} = \sin^{-1} \frac{x}{3}$

④ cont. $A = -6 \left[\cos \frac{\pi}{2} - \cos 0 \right] = -6(-1) = 6$ (3)

⑤ a) $\cos^{-1} \left(-\frac{1}{2} \right) = \pi - \cos^{-1} \frac{1}{2} = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$ (2)

b) $\tan^{-1} \left(\tan \frac{2\pi}{3} \right) = \tan^{-1} \left(-\frac{\sqrt{3}}{3} \right) = -\tan^{-1} \frac{\sqrt{3}}{3} = -\frac{\pi}{3}$ (2)

c) $\sin \left[\tan^{-1} \left(-\frac{2}{3} \right) + \cos^{-1} \left(\frac{12}{13} \right) \right] = \sin \left[-\tan^{-1} \left(\frac{2}{3} \right) + \cos^{-1} \left(\frac{12}{13} \right) \right]$

$= \sin(\beta - \alpha) = \sin \beta \cos \alpha - \cos \beta \sin \alpha$

$= \frac{5}{13} \cdot \frac{3}{\sqrt{13}} - \frac{12}{13} \cdot \frac{2}{\sqrt{13}} = \frac{-9}{13\sqrt{13}}$ (3)

⑥ $\int_0^{\pi/2} \sqrt{4-x^2} dx$ $x = 2 \sin \theta$ $x=0 \Rightarrow \theta=0$
 $0 = 2 \sin \theta$
 $\frac{dx}{d\theta} = 2 \cos \theta$ $x=2 = 2 \sin \theta$
 $1 = \sin \theta$
 $\theta = \pi/2$
 $dx = 2 \cos \theta \cdot d\theta$ $x = \pi/2$ (1)

$= \int_0^{\pi/2} 2 \sqrt{1-\sin^2 \theta} \cdot 2 \cos \theta d\theta$

$= \int_0^{\pi/2} 4 \cos^2 \theta d\theta$ $\cos 2\theta = 2 \cos^2 \theta - 1$ (4)
 $\frac{1}{2}(\cos 2\theta + 1) = \cos^2 \theta$

$= \int_0^{\pi/2} 4 \cos^2 \theta d\theta = 2 \int_0^{\pi/2} (\cos 2\theta + 1) d\theta = 2 \left[\frac{1}{2} \sin 2\theta + \theta \right]_0^{\pi/2}$

$= \left[\sin 2\theta + 2\theta \right]_0^{\pi/2} = 0 + \pi - 0 - 0 = \pi$ (2)

$$\begin{aligned} \textcircled{7} \int \frac{(\ln x)^2}{x} dx & \quad u = \ln x \\ \frac{du}{dx} &= \frac{1}{x} \quad \therefore x du = dx \\ &= \int \frac{u^2}{x} \cdot x du \\ &= \int u^2 du = \frac{u^3}{3} + C = \frac{(\ln x)^3}{3} + C \end{aligned}$$

$$\textcircled{8} v^2 = 16x - 4x^2 + 20$$

$$a) \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \ddot{x}$$

$$\text{since } v^2 = 16x - 4x^2 + 20$$

$$\therefore \frac{1}{2} v^2 = 8x - 2x^2 + 10$$

$$\therefore \ddot{x} = \frac{d}{dx} (8x - 2x^2 + 10) = 8 - 4x \quad \textcircled{3}$$

$$\therefore \ddot{x} = -4(x-2) \text{ which is in the form}$$

$$\ddot{x} = -n^2(x-m) \quad \text{where } n=2$$

$\therefore$ SHM

centre of the motion $x=m=2$

$$b) v=0 = 16x - 4x^2 + 20$$

$$0 = 4(4x - x^2 + 5)$$

$$0 = -(x^2 - 4x - 5) = -(x-5)(x+1)$$

$$x = -1 \quad x = 2 \quad x = 5 \quad \text{extremities}$$

$$c) T = \frac{2\pi}{n} = \frac{2\pi}{2} = \pi \quad \textcircled{1}$$

⑧ cont.

d) speed_{max} is at $x=2$ (centre)

$$\therefore v^2 = 16 \times 2 - 4 \times 2^2 + 20 = 36$$

$$\therefore |v| = \text{speed} = \sqrt{36} = 6 \text{ ms}^{-1}$$

$$\textcircled{9} t=0 \quad x=+2 \text{ m} \quad v=5 \text{ ms}^{-1}$$

$$\ddot{x} = a = 2x^3 + 2x$$

$$a) \ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 2x^3 + 2x$$

$$\frac{1}{2} v^2 = \int 2x^3 + 2x dx$$

$$\frac{1}{2} v^2 = \frac{x^4}{2} + x^2 + C$$

$$x=2 \quad v=5$$

$$\frac{1}{2} \times 25 = \frac{2^4}{2} + 2^2 + C$$

$$\frac{1}{2} = C$$

$$\therefore \frac{1}{2} v^2 = \frac{x^4}{2} + x^2 + \frac{1}{2} \therefore v^2 = x^4 + 2x^2 + 1$$

$$\therefore v^2 = (x^2 + 1)^2$$

$$\therefore v = \pm (x^2 + 1) \text{ but } t=0 \quad v=+5$$

$$\therefore v = \oplus (x^2 + 1) \text{ and } v \neq 0 \quad \left[\begin{array}{l} \text{justify } \oplus \text{ only} \\ v = x^2 + 1 \text{ always} \end{array} \right]$$

only $\rightarrow$ shown

$$b) v = \frac{dx}{dt} = x^2 + 1 \quad \therefore \frac{dx}{x^2 + 1} = dt \quad \therefore \int \frac{1}{x^2 + 1} dx = \int dt$$

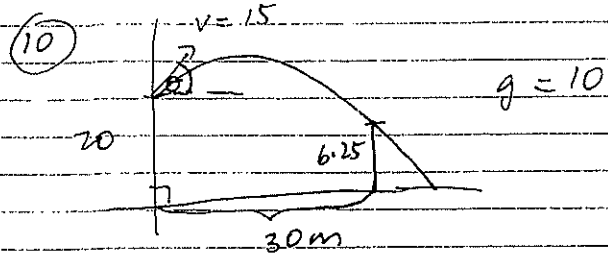
$$\therefore \frac{dt}{dx} = \frac{1}{x^2 + 1} \therefore t = \int \frac{1}{x^2 + 1} dx$$

$$t=0 \quad x=2$$

$$t = \tan^{-1} x + C$$

$$\therefore 0 = \tan^{-1} 2 + C \quad \therefore C = -\tan^{-1} 2 \quad \therefore t = \tan^{-1} x - \tan^{-1} 2$$

-5-



(a) $\ddot{x} = 0$ $\ddot{y} = -g$
 $\dot{x} = \int 0 dt = C = v \cos \theta$ $\dot{y} = \int -g dt$
 $x = \int v \cos \theta dt = vt \cos \theta + C$ $y = -gt + C$
 $t=0, x=0=C$ $t=0, y=20=C$
 $\therefore x = vt \cos \theta$ $\therefore y = -\frac{1}{2}gt^2 + vt \sin \theta + 20$
 $\therefore x = 15t \cos \theta$ $t=0, y=20=C$
 $\therefore y = -5t^2 + 15t \sin \theta + 20$

(b) (1) $t = \frac{x}{15 \cos \theta}$
 (2) $y = -5 \frac{x^2}{15^2 \cos^2 \theta} + 15 \frac{x}{15 \cos \theta} \sin \theta + 20$
 $y = -\frac{5x^2}{15^2} (1 + \tan^2 \theta) + x \tan \theta + 20$
 $y = -\frac{1}{45} x^2 (1 + \tan^2 \theta) + x \tan \theta + 20$
 $6.25 = -\frac{1}{45} \times 30^2 (1 + \tan^2 \theta) + 30 \tan \theta + 20$
 $6.25 = -20 - 20 \tan^2 \theta + 30 \tan \theta + 20$
 $20 \tan^2 \theta - 30 \tan \theta + 6.25 = 0$

-6-

$\therefore \tan \theta = \frac{30 \pm \sqrt{30^2 - 4 \times 20 \times 6.25}}{2 \times 20} = \frac{30 \pm 20}{40} = \frac{5}{4}$
 $\therefore \theta = 14^\circ 2' \text{ or } 51^\circ 20'$

c) $t = \frac{x}{15 \cos \theta}$

$t = \frac{x=30}{15 \times \frac{4}{\sqrt{17}}} = \frac{30 \sqrt{17}}{15 \times 4}$

$\therefore t = \frac{\sqrt{17}}{2}$

smaller
 $\tan \theta = \frac{1}{4}$

$\therefore \cos \theta = \frac{4}{\sqrt{17}}$

(2) Total 51